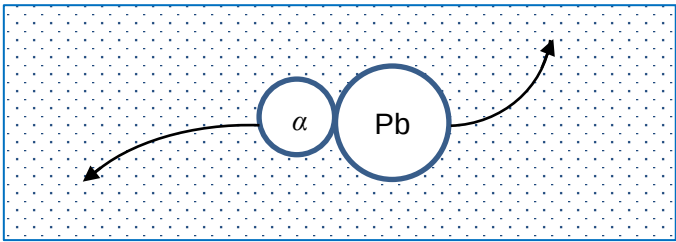


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(a)	The half-life of a radioactive nuclide is the mean time taken for a quantity x to reduce to half its initial value where x represents either the number, activity or count rate of radioactive nuclei.
(b)	Spontaneous: Not affected by chemical reactions and physical conditions such as temperature and pressure.
	Random: No way to predict which particular nucleus in a radioactive sample will decay next. Each nucleus in a sample has the same chance of decaying per unit time.
(c)(i)	${}^{14}_6\text{C} \rightarrow {}^{14}_7\text{N} + {}^0_{-1}\text{e} + \text{energy}$
(c)(ii)	$\lambda = \frac{\ln 2}{5730 \times 365 \times 24 \times 60 \times 60}$
	$= 3.84 \times 10^{-12} \text{ s}^{-1} \text{ (3 s.f.)}$
(c)(iii)	$A = A_0 e^{-\lambda t}$
	$5.4 = (0.23 \times 10000) e^{-3.84 \times 10^{-12} t}$
	$t = 1.577 \times 10^{12} \text{ s}$
	$t = \frac{1.577 \times 10^{12}}{365 \times 24 \times 60 \times 60}$ $= 50000 \text{ years (3 s.f.)}$
(d)(i)	$m_{\text{Pb}} u_{\text{Pb}} + m_{\alpha} u_{\alpha} = m_{\text{Pb}} v_{\text{Pb}} + m_{\alpha} v_{\alpha}$ $0 = 206 v_{\text{Pb}} + 4 v_{\alpha}$

	$KE = \frac{p^2}{2m}$ $\frac{KE \text{ of Pb}}{KE \text{ of } \alpha} = \frac{\frac{p_{Pb}^2}{2m_{Pb}}}{\frac{p_{\alpha}^2}{2m_{\alpha}}} = \frac{m_{\alpha}}{m_{Pb}} = \frac{4}{206}$ $= 0.019417$ $= 0.0194 \text{ (3 s.f.) (shown)}$
(d)(ii)	 Magnetic force provides centripetal force $Bqv = \frac{mv^2}{r}$ $r = \frac{mv}{Bq}$ • both nuclei have the same momentum • Pb nucleus has much larger charge than α-particle • Pb nucleus has smaller radius than α-particle • apply Fleming's left hand rule to determine direction of motion Opposite direction for Pb nucleus (right) and α-particle (left) Correct path for Pb nucleus and α-particle with smaller radius for Pb nucleus
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